

YI DONG

FPGA / RTL / Digital Design Intern

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Engineering Page: dydx404.github.io/cv

Education

Imperial College London

London, UK

MEng Electronic and Information Engineering

Expected graduation: 2028

- Relevant coursework: Digital Electronics, Computer Architecture, Signals and Systems, Communications, Control Systems, Embedded Systems, Mathematics for Engineers, Software Systems.

Skills

FPGA / Digital Design: SystemVerilog, Verilog, FSMs, pipelined RTL datapaths, Q-format fixed point, CORDIC, stencil/FDTD compute, PML, CDC/async-FIFO, BRAM ping-pong buffering, timing closure, Vivado

SoC / Interfaces: Zynq-7000, PS/PL partitioning, AXI4-Stream/Lite, AXI DMA, AXI GPIO/MMIO, HDMI/DVI (rgb2dvi), PYNQ overlays, pipelined RV32I CPU

Embedded / Systems: C/C++, Python, Bash, Linux, OpenCV, UDP sockets, PYNQ-Z1, ESP32

Verification / Tooling: Verilator, Icarus Verilog, Vivado/XSim, waveform debugging, directed tests, Python reference models, Git

Projects

Real-Time Electromagnetic FDTD Accelerator with 3D HDMI Renderer

Feb 2026–Jun 2026

SystemVerilog, Zynq-7020/PYNQ-Z1, Q3.13 fixed point, CORDIC, PML, BRAM, AXI GPIO, HDMI

GitHub

- Built a hardware-validated 2D FDTD electromagnetic solver on PYNQ-Z1 (Q3.13 datapaths, CORDIC source, PML boundaries, ping-pong BRAM buffers), scaled from 64×64 to a 128×128 four-lane engine.
- Integrated a 3D ray-march renderer (100 MHz core, async-FIFO to a 25 MHz pixel domain) driving a live propagating EM wave to HDMI via rgb2dvi.
- Exposed PS-side runtime control over AXI GPIO: PYNQ/Python streams Q3.13 camera vectors (yaw/pitch/distance) into the PL, plus live source, display-mode, and height-scale tuning without rebuilds.
- Closed timing at 100 MHz (WNS +0.393 ns, BRAM 105/140, DSP 179/220); debugged a free-run energy runaway (soft source), a silent AXI address-map reorder bug, and a power-of-two march-step DSP blowup.

FPGA Wand / PYNQ-Z1 Vision System

Feb 2026–Mar 2026

PYNQ-Z1, Zynq-7000, Verilog, AXI DMA, AXI-Stream, AXI-Lite/MMIO, OpenCV, UDP

GitHub

- Built a PYNQ-Z1 vision prototype integrating OpenCV camera capture, PS-side preprocessing, AXI DMA transfer, FPGA-side centroid extraction, and UDP point streaming.
- Implemented an AXI4-Stream RTL module for thresholded-frame centroid accumulation, exposing sum, count, frame_id, and valid through AXI-Lite/MMIO to offload per-frame PS work.
- Brought up the system on real hardware; debugged camera init, DMA behavior, overlay/HWH mismatches, duplicate frames, and noise-sensitive thresholds.

RV32I Pipelined CPU with Cache

Nov 2025–Dec 2025

SystemVerilog, RV32I pipeline, cache, hazards, forwarding, Verilator

GitHub

- Implemented and tested a simplified RV32I five-stage pipelined CPU in SystemVerilog as a teaching-oriented computer architecture project.
- Integrated forwarding paths, load-use hazard handling, stall/flush control, writeback selection, byte-addressable memory, and an experimental cache/refill path.
- Used Verilator/GoogleTest and directed RV32I assembly/reference programs to check load/store behavior, forwarding cases, pipeline control, and writeback results.
- Debugged PC flow, branch flushes, pipeline stalls, a0/writeback visibility, and cache interactions using VCD waveforms and text traces.

GTX High-Speed Serial Transceiver Loopback (Self-Directed Exploration)

Jun 2026–Present

Zynq-7100, Kintex-7 GTX, 3.125 Gb/s, 8b/10b, CPLL, near-end PCS loopback, Vivado

- Self-directed study of Xilinx GTX high-speed serial transceivers, working through architecture, clocking, reset sequencing, and 8b/10b framing toward a simulation-only near-end PCS loopback on Zynq-7100.
- Captured the design bottom-up in a decision log and verification plan, including a bring-up FSM for byte-alignment stability; formalizing the checker FSM ahead of RTL.